Note in Optimization Theory

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Abstract

The note is about something about the algorithm-the course's primary reference [1]. Thanks to my teacher Bo Jiang.

The main contents are organized as follows. In section 1, we discuss the optimality condition from unconstraint to inequality constraint. Here, a trick of proof is about the penalty function regarding constraint's square and the distance as penalty term. In section2, we discuss some method to determine step size or learning rate, including exact minimization(steepest gradient descent, Armijio's rule, diminishing step, and constant step)

Keywords: Optimality Condition, KKT, first-order algorithm, steepest gradient descent, Armijio's rule, diminishing step, constant step

1 Optimality Condition

Consider the nonlinear programming with explicit equality and inequality constraints:

$$\begin{aligned} minimize \quad & f(x) \\ s.t. \quad & h_i(x) = 0, i = 1, \cdots, m, \\ & g_j(x) \leq 0, j = 1, \cdots, r \end{aligned} \tag{1}$$

Our goal: to characterize the optimal solution of the problem above.

1.1 consider an easy case: unconstrained problem

$$\min_{x \in \mathbb{R}^n} f(x)$$

First-order condition

By **Fermat's rule**(**The necessity of local optimum**), we have

$$\nabla f(x) = [\frac{\partial f(x)}{\partial x_1}, \cdots, \frac{\partial f(x)}{\partial x_n}]^T = 0$$

Maybe this point is a saddle point, so we need to calculate its second-order condition.

Second-order condition

The Necessary Condition is

$$\nabla^2 f(x) \geq 0$$

A straight line parallel to the x-axis exists a local minimum point. Its second-order condition is 0.

The Sufficient Condition is

$$\nabla^2 f(x) \succ 0$$

1.2 If the problem with convex set constraint is Non-differentiable

$$\min_{x \in S} f(x)$$

where S is a closed convex constraint.

Necessary optimality condition:

$$\nabla f(x)^T (y - x) \geq 0, \forall y \in S$$

Proof. Suppose x is a local minimum $f(y) = f(x) + \nabla f(x)^T (y - x) + O(\|y - x\|)$ Suppose there exists a $y \in S$, $\nabla f(x)^T (y - x) < 0$ denote $d = y - x$,

$$z = x + \alpha d, (0 \leq \alpha \leq 1)$$

when $\alpha \to 0$, $z \to x$

$$f(z) = f(x) + \nabla f(x)^T (z - x) + O(\|z - x\|) = f(x) + \nabla f(x)^T (\alpha d) + O(\|\alpha d\|)$$

which is a contradiction to local minimal. □

The proof idea is if the opposite condition holds, then there is a point used by linear approximation smaller than the initial point, which is contradictory to it is a local minimum.

1.3 Problem with only equality constraints

$$\begin{aligned} min \quad & f(x) \\ s.t. \quad & h_i(x) = 0, i \ldots, m \end{aligned} \tag{2}$$

Theorem 1.1. *Suppose that the constraints are regular i.e.*

$$rank([\nabla h_1(x), \ldots, \nabla h_m(x)]) = m$$

for any x. If x^* is an optimal solution, then there must exist $\lambda_i, i, \ldots, m$ such that

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla h_i(x^*) = 0$$

1.3.1 second-order Sufficient Condition (Optimality)

For any $y \neq 0$ and $H(x^*)^T y = 0$, the x^* satisfies

$$y^\top \left[\nabla^2 f\left(x^*\right) + \sum_{i=1}^{m} \lambda_i \nabla^2 h_i\left(x^*\right) \right] y > 0$$

then x^* is a local minimum.

1.3.2 second-order Necessity Condition (Optimality)

$$\begin{aligned} \nabla_{xx} L(x^*, \lambda) &\geq 0 \\ \nabla^2 f(x) + \sum_{i=1}^{m} \lambda_i \nabla^2 h(x) &\geq 0 \end{aligned} \tag{3}$$

don't need for any y.

Sketch of the proof. Consider

$$F_\mu(x) = f(x) + \frac{\mu}{2} \sum_{i=1}^{m} h_i^2(x) + \frac{\nu}{2} \|x - x^*\|^2$$

which is named penalty function where the second term is used to penalize violating constraint, and the last term is used to penalize the distance to optimal solution.

$$\begin{aligned} 0 = \nabla F_{\mu_k}(x^k) =& \nabla f(x^k) + \mu_k \sum_{i=1}^{m} h_i(x^k)[\nabla h_i(x^k)] + \nu(x^k - x^*) \\ \lim_{k\to\infty} 0 =& \nabla f(x^k) + \lim_{k\to\infty} \mu_k \sum_{i=1}^{m} h_i(x^k) \nabla h_i(x^k) \end{aligned} \tag{4}$$

Denote

$$\begin{aligned} H(x) &= [\nabla h_1(x), \ldots, \nabla h_m(x)] \\ h(x) &= [h_1(x), \ldots, h_m(x)]^T \end{aligned} \tag{5}$$

So by these two notations, we have

$$0 = \nabla F_{\mu_k}(x^k) = \nabla f(x^k) + \nu(x^k - x^*) + \mu_k H(x^k) h(x^k) \tag{6}$$

So the natural idea is $y = H^{-1}(x_k)z$ But $H(x_k) \in R^{n\times m}$ maybe not have inverse matrix.

Key: multiply both side of the equation by $H(x_k)^T$, then $H(x_k)^T H(x_k)$ is a $m \times m$ matrix, and it is inverse.

Then, we have

$$\mu_k h(x^k) = -(H(x^k))$$

$$\mu_k h(x_k) = -(H(x_k)^T H(x_k))^{-1}(\nabla f(x_k) + \nu(x^k - x^*))$$

equate λ to $\lim_{k\to\infty} \mu_k h(x^k)$

This gives us the relation.

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla h_i(x^*) = 0$$

Now we prove the first-order condition successfully. The second-order necessity condition is manageable. □

1.4 Problem with equality and inequality constraints

$$\begin{aligned} min \quad & f(x) \\ s.t. \quad & h_i(x) = 0, i = 0 \ldots, m \\ & g_j(x) \leq 0, j = 0 \ldots, r \end{aligned} \tag{7}$$

Definition 1.1 (active constraint)**.** Let $A(x) = \{j | g_j(x) = 0\}$.

1.4.1 Sufficient condition

Suppose that the constraints are regular,i.e.[1]

$$\nabla h_1(x), \cdots, \nabla h_m(x), \nabla g_j(x), j \in A(x)$$

are linear independent for any given x. Then if x^* is an optimal solution, there must exist $\lambda_i, i = 1, \ldots, m, \mu_j \geq 0, j = 1, \ldots, r$, such that

$$\nabla_x L(x^*, \lambda, \mu) = 0 \quad \text{and} \quad y^\top \nabla_x^2 L(x^*, \lambda, \mu) y > 0$$

for all y such that $\nabla h_i(x^*)^T y = 0, i = 0, \ldots, m, \nabla g_j(x^*)^T y = 0, j \in A(x^*)$. Moreover, $\mu_j = 0$ for all $j \neq A(x^*)$(KKT condition)

Proof. We may convert an inequality constraint into an equality constraint, resulting in an equivalent problem.

$$\begin{aligned} \text{minimize} \quad & f(x) \\ \text{subject to} \quad & h_i(x) = 0, i = 1, \cdots, m \\ & g_j(x) + z_j^2 = 0, j = 1, \cdots, r \end{aligned}$$

Let the Lagrangian multipliers be λ (first m constraints) and μ (next r constraints).

The Lagrangian function is $L(x, z, \lambda, \mu)$, moreover

$$\nabla L(x, z, \lambda, \mu) = \begin{bmatrix} \nabla f(x) + \sum_{i=1}^{m} \lambda_i \nabla h_i(x) + \sum_{j=1}^{r} \mu_j \nabla g_j(x) \\ 2\mu_1 z_1 \\ \vdots \\ 2\mu_r z_r \end{bmatrix}$$

[1]the constraint had better not too serious like don't writing it as quadratic function when it's a linear constrain.

The first-order Optimality condition gives (refer to equality Optimality condition)

$$\begin{cases} \nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla h_i(x^*) + \sum_{j=1}^r \mu_j \nabla g_j(x^*) = 0 \\ 2\mu_j z_j^* = 0, j = 1, \ldots, r \end{cases} \tag{8}$$

ItsHessian matrix is

$$\nabla^2 L_{x,z}(x, z, \lambda, \mu) = \begin{bmatrix} \nabla^2 f(x) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(x) + \sum_{j=1}^r \mu_j \nabla^2 g_j(x) & 0 \\ 0 & 2\,\mathrm{diag}(\mu) \end{bmatrix}$$

Hence

$$\begin{aligned} &\begin{bmatrix} y \\ w \end{bmatrix}^\top \nabla^2_{x,z} L(x, z, \lambda, \mu) \begin{bmatrix} y \\ w \end{bmatrix}^\top \\ &= y^\top \left[\nabla^2 f(x) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(x) + \sum_{j=1}^r \mu_j \nabla^2 g_j(x) \right] y + 2 \sum_{j=1}^r \mu_j w_j^2 \end{aligned}$$

The second-order optimality condition states that

$$\begin{bmatrix} y \\ w \end{bmatrix}^\top \nabla^2_{x,z} L(x, z, \lambda, \mu) \begin{bmatrix} y \\ w \end{bmatrix}^\top \geq 0$$

for all

$$\left(\nabla_{x,z} h_i(x, z)\right)^\top \begin{bmatrix} y \\ w \end{bmatrix} = 0 \text{ and } \left(\nabla_{x,z} g_j(x, z)\right)^\top \begin{bmatrix} y \\ w \end{bmatrix} = 0$$

Now $\nabla_{x,z} h_i(x, z) = \begin{bmatrix} \nabla h_i(x) \\ 0 \end{bmatrix}$ and $\nabla_{x,z} g_j(x, z) = \begin{bmatrix} \nabla g_j(x) \\ 2z_j \end{bmatrix}$. Therefore,

$$\left(\nabla_{x,z} g_j(x, z)\right)^\top \begin{bmatrix} y \\ w \end{bmatrix} = \left(\nabla g_j(x)\right)^\top y + 2 z_j w_j$$

□

The proof's idea is to add some terms to make the inequality change to equality, then using the conclusion be proven by equality constraint.

1.4.2 Without any regularity condition

Consider the following example:

$$\begin{aligned} \text{minimize} \quad & x_1^2 + x_2^2 \\ \text{subject to} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ & (x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \end{aligned}$$

in the example, the two gradients are Linear correlation, so the KKT condition here is invalid.

So you can not find multipliers to make the optimality condition hold.

1.5 Fritz John condition

However, the following condition always holds:

There are $\lambda_i, i = 1, \cdots, m, \mu_i \geq 0, j = 0, 1 \cdots, r$

$$\mu_0 \nabla f\left(x^*\right) + \sum_{i=1}^{m} \lambda_i \nabla h_i\left(x^*\right) + \sum_{j=1}^{r} \mu_j \nabla g_j\left(x^*\right) = 0$$

where $\mu_j g_j(x^*) = 0, j = 1, \ldots, r$, and not all these multipliers are zero.

REMARK: if $\mu_0 > 0$, then the condition is equivalent to KKT condtion; if $\mu_0 = 0$, it can't help to get solution, so how to find other qualification to help us find optimal solution

1.5.1 Mangasarian-Fromovitz constraint

$\nabla h_i(x^*), i = 1, \ldots, m$ are linearly independent, and there exists d such that

$$(\nabla h_i(x^*))^T d = 0, \forall i \tag{9}$$

and

$$(\nabla g_j(x^*))^T d < 0 \forall j \in A(x^*) \tag{10}$$

It is easy to see that with this condition, one must have $\mu_0 > 0$

Proof. Fritz John condition remind us not all these multiplier(μ_j) are zero. By first-order condition

$$\mu_0 \nabla f(x) + \sum_{i=1}^{m} \lambda_i \nabla h_i(x) + \sum_{j=1}^{r} \mu_j \nabla g_j(x) = 0$$

Suppose $\mu_0 = 0$ Then

$$\left[\sum_{i=1}^{m} \lambda_i \nabla h_i(x) + \sum_{j=1}^{r} \mu_j \nabla g_j(x)\right]^T d = 0$$

by 9 and 10,we have $\mu_j = 0 \forall j$ Subtitute $\mu_j = 0$ into Fritz John condition, then

$$\sum_i \lambda_i \nabla h_i(x) = 0 \Rightarrow \lambda_i = 0$$

which is on the contrary to existing $\mu_0 \neq 0$ □

1.5.2 Slater constraint qualification

This is a very well know and simple, verifiable constraint qualification if all $h_i's$ are linear, and $g_j's$ are convex.

There is $\hat{x}$ such that $h_i(\hat{x}) = 0 (\forall i = 1, \ldots, m)$ and $g_j(\hat{x}) < 0 (\forall j = 1 \ldots, r)$

Proof.

$$0 > g_j(\hat{x}) \geq \bar{g}_j\left(x^*\right) + \left(\nabla g_j\left(x^*\right)\right)^\top (\hat{x} - x^*) = \left(\nabla g_j\left(x^*\right)\right)^\top (\hat{x} - x^*)$$

The second inequality is by convex function's property, and the equality is by $j \in A(x^*), g_j(x^*) = 0$ for $j \in A(x^*)$. Therefore, if we let $d = \hat{x} - x^*$, then

$$(\nabla g_j(x^*))^T d < 0$$

for all $j \in A(x^*)$ and

$$(\nabla h_i(x^*))^T d = 0$$

due to the linearity of $h_i, i = 1, \ldots, m$ □

2 How to determine step size

2.1 Exact minimization:

Find $\alpha_k > 0$ such that

$$f(x^k + \alpha_k d^k) = \min_{\alpha \geq 0} f(x^k + \alpha d^k)$$

can be implemented by line search

1. bi-section
2. golden-section: Fibonacci

2.2 Armijio's Rule

Fix $s > 0, \beta \in (0, 1)$ and $\sigma \in (0, 1)$. Let $m > 0$ be the smallest integer satisfying

$$f(x^k) - f(x^k + \beta^m s d^k) \geq -\sigma \beta^m s \nabla f(x^k)^T d^k$$

and the stepsize $\alpha_k = s\beta^m$

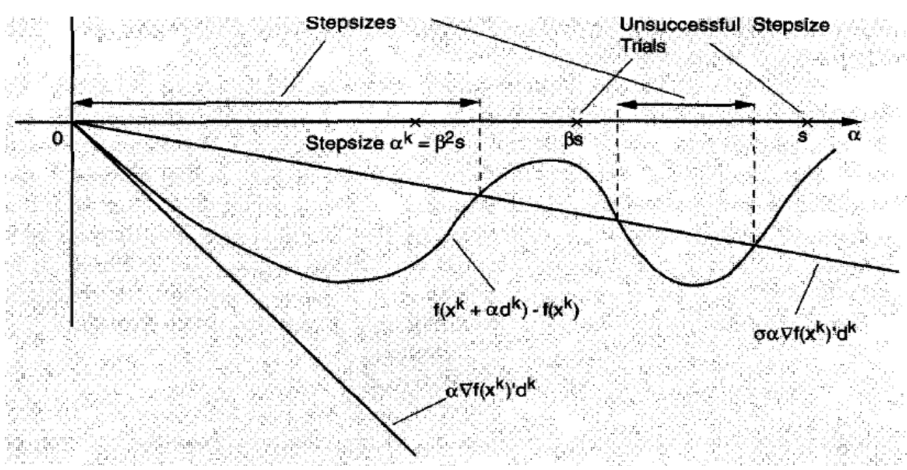

Figure 1: Armijio

2.3 diminishing step

$\alpha_k \downarrow 0$ and $\sum_{k=1}^{\infty} \alpha_k = \infty$

3 Basic concepts of convergence

3.1 Possible error measurements

1. $e(x^k) = \left\|x^k - x^*\right\|$
2. $f(x^k) - f(x^*)$
3. $\left\|\nabla f(x^k)\right\|$

Criterion 1 and 2 are always used in theoretical analysis, and Criterion 3 is mostly used in algorithm practice.

3.2 Two methods to measure convergence velocity

1. iteration complexity: $f(x^k) - f(x^*) \leq O(\frac{1}{k})(O(\frac{1}{k^2}), O(\frac{1}{k^3}), \cdots)$
2. $$\lim_{k\to\infty} \frac{e\left(x^{k+1}\right)}{e\left(x^k\right)} = \mu$$

   where $e\left(x^k\right)$ could be $\left\|x^k - x^*\right\|, \|f\left(x^k\right) - f\left(x^*\right)\|, \ldots$

   To distinguish superlinear rates of convergence, we check
   $$\lim_{k\to\infty} \frac{e(x^{k+1})}{e(x^k)^q} = \mu > 0$$

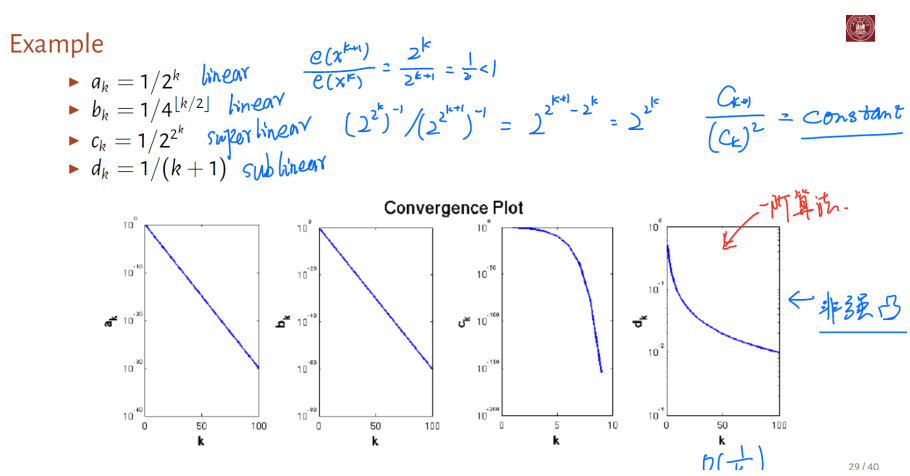

Figure 2: Convergence plot

4 GD convergence analysis for strongly convex function

By the mean-value theorem we have

$$f(y+\alpha d) = f(y) + \alpha \nabla f(y)^\top d + \frac{\alpha^2}{2} d^\top \nabla^2 f(\bar{y}) d + O(\alpha^2 \|d\|^2)$$

where $\bar{y}$ lies between y and $y+\alpha d$, $\exists M \geq m \geq 0 s.t.$

$$\begin{aligned} & MI \geq \nabla^2 f(x) \geq mI \\ & My^T y \geq y^T \nabla^2 f(x) y \geq m y^T y \\ & f(y) + \alpha \nabla f(y)^T d + \frac{\alpha^2 m}{2} \|d\|^2 \leq f(y+\alpha d) \leq f(y) + \alpha \nabla f(y)^T d + \frac{\alpha^2 M}{2} \|d\|^2 \end{aligned} \tag{11}$$

If we let $\alpha = \frac{1}{M}$ and $d = -\nabla f(x)$, then

$$f(x+\alpha d) \leq f(x) - \frac{1}{2M} \|\nabla f(x)\|^2 \tag{12}$$

On the other hand, let $y+\alpha d = x^*$,$\alpha d = x^* - y$, substitute into last left term in 11, then we have

$$f(x) - f(x^*) \leq \|\nabla f(x)\| \cdot \|x - x^*\| - \frac{m}{2} \|x - x^*\|^2 \leq \frac{1}{2m} \|\nabla f(x)\|^2 \tag{13}$$

The first inequality generated by cauchy-schwarz inequality and the second inequality by using quadratic function.

Substituting this into an early inequality yields

$$f(x+\alpha d) \leq f(x) - \frac{1}{2M} \|\nabla f(x)\|^2 \leq f(x) - \frac{m}{M} (f(x) - f(x^*)) \tag{14}$$

In otherwords,we have the linear convergent rate

$$f(x+\alpha d) - f(x^*) \leq \left(1 - \frac{m}{M}\right) (f(x) - f(x^*)) \tag{15}$$

It is possible to improve the rate a little bit. In fact, one can improve the rate to (no proof)

$$f(x+\alpha d) - f(x^*) \leq \left(\frac{1 - \frac{m}{M}}{1 + \frac{m}{M}}\right)^2 (f(x) - f(x^*)) \tag{16}$$

REMARK: utilize the second-order condition to bound the Hessian matrix. And use Taylor's Formula second-order expansion to bound the next iteration's value11. We are using Cauchy-Schwarz inequality to transform negative term to positive term14.

5 Convergence in more general setting

Definition 5.1 (Lipschitz constant)**.** Lipschitz continuous of gradient

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|$$

Lemma 5.1 (Descent Lemma)**.** *If the Lipschitz continuous gradient is satisfied, we have*

$$f(y) - f(x) \leq \nabla f(x)^T (y - x) + \frac{L}{2} \|y - x\|^2$$

Proof. Consider $g(t) = f(x + t(y - x))$, then $g(0) = f(x)$ and $g(1) = f(y)$, then

$$\begin{aligned} f(y) - f(x) &= \int_0^1 g'(t)dt = \int_0^1 < y - x, f(x + t(y - x)) > dt \\ &= < \nabla f(x), y - x > + \int_0^1 < \nabla f(x + t(y - x)) - \nabla f(x), y - x > dt \\ &\leq < \nabla f(x), y - x > + \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\|_2 \|y - x\|_2 \, dt \\ &\leq \frac{L}{2} \|y - x\|_2^2 \end{aligned} \tag{17}$$

□

Remark: The idea of proof is transform $f(y) - f(x)$ what we want to prove to an integral. And then using linear function $< \nabla f(x), y - x >$ to approximate primal integral, the last step is using Lipschitz constant to bound left integral.

6 Accelerated Gradient Descent

1. GD focuses on improving the cost per iteration, which might sometimes be too "short-sighted"
2. GD might sometimes zigzag or experience abrupt changes

6.1 Heavy ball method

1. Exploit information from the history (i.e. past iterates)
2. Add buffers (like momentum) to yield smoother trajectory

$$\boldsymbol{x}^{t+1} = \boldsymbol{x}^t - \eta_t \nabla f\left(\boldsymbol{x}^t\right) + \underbrace{\theta_t \left(\boldsymbol{x}^t - \boldsymbol{x}^{t-1}\right)}_{\text{momentum term}} \tag{18}$$

6.2 Nesterov's algorithm

$$\begin{aligned} x^{(t+1)} &= y^{(t)} - \eta_t \nabla f\left(y^t\right) \\ y^{t+1} &= x^{t+1} + \tfrac{t}{t+3}\left(x^{t+1} - x^t\right) \end{aligned} \tag{19}$$

The idea of Nesterov's algorithm is to use two different iterative sequences to correct each other. (issue: why not correct sequence x after updating sequence x)

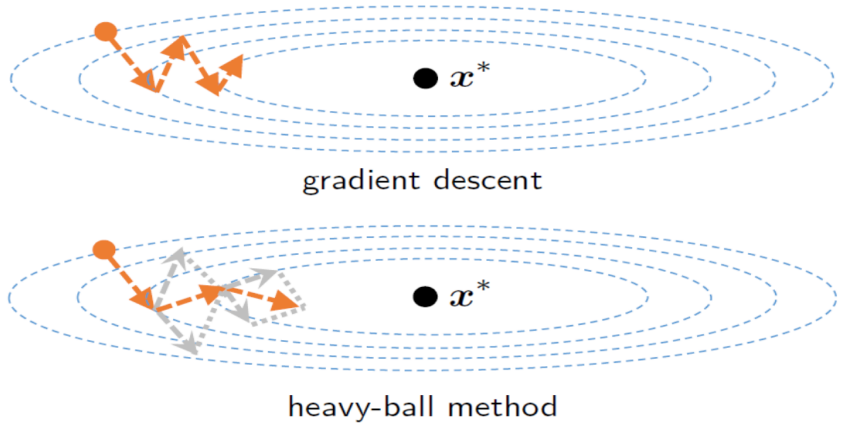

Figure 3: Heavy ball

7 structured nonsmooth problem

Unconstrained optimization with objective split in two components:

$$\min f(x) = g(x) + h(x)$$

g is convex differentiable, h is convex but non-differentiable.

Example 7.1. Lasso problem:

$$\min_x \tau \|x\|_1 + \frac{1}{2}\|Ax - b\|_2^2$$

Example 7.2. General matrix completion: We hope only low-rank matrix consistent with the sampled entries.

Consider the rank minimization problem:

$$\begin{aligned} \min_x \quad & rank(X) \\ s.t. \quad & X_{ij} = M_{ij} \end{aligned} \tag{20}$$

this is a N-p hard problem, and now we want to find a convex function to approximate its rank, $rank(x) = \|\lambda(x)\|_0 \approx \|\lambda(x)\|_1 = \sum_{i=1}^n |\lambda_i(x)| = \|x\|_*$

Definition 7.1 (Nuclear norm)**.** singular value decomposition

$$X = \sum_{k=1}^{r} \sigma_k u^k v^{k^T}$$

Nuclear norm ($\sigma_i(x)$ is the i-th largest singular value of X):

$$\|X\|_* = \sum_{i=1}^{n} \sigma_i(X)$$

Nuclear norm minimization:

$$\begin{aligned} \min \quad & \|X\|_* \\ \text{s.t.} \quad & \mathcal{A}(X) = b \end{aligned} \tag{21}$$

where $\mathcal{A}$ is a linear mapping of X.

7.1 Main idea

Unconstrined optimization with objective split in two components:

$$\min f(x) = g(x) + h(x)$$

Linearize the smooth part and look at the following subproblem:

$$\min_y g(x^{k-1}) + \nabla g(x^{k-1})^T (y - x^{k-1}) + h(y) + \frac{t_k}{2} \left\| y - x^{k-1} \right\|^2$$

or equivalently,

$$\min_y \frac{1}{2t_k} \left\| y - (x^{k-1} - t_k \nabla g(x^{k-1})) \right\|^2 + h(y)$$

Next I want to talk about why is this sub-problem.

7.2 sub-problem and Iterative format

Example 7.3.

$$\begin{aligned} x^{k+1} =& x^k - \alpha \nabla f(x^k) \\ & \min_x f(x^k) + \nabla f(x^k)^T (x - x^k) + \frac{1}{2\alpha} \|x - x_k\|^2 \end{aligned} \tag{22}$$

7.3 Proxmial mapping

The proximal mapping (prox-operator) of a convex function h is defined as

$$\text{prox}_h(x) = \arg\min_y \left(h(y) + \frac{1}{2} \|y - x\|_2^2 \right) \tag{23}$$

Remark: The idea of proximal mapping is put **distance** as penalty term.

7.4 Algorithm (ISTA)(Iterative Shrinkage-Thresholding Algorithm)

$$x^k = \arg\min_x \frac{1}{2t_k} \left\| x - \left(x^{k-1} - t_k \nabla g \left(x^{k-1} \right) \right) \right\|^2 + h(x) \tag{24}$$

Or equivalently,

$$x^k = \text{prox}_{t_k h} \left(x^{k-1} - t_k \nabla g \left(x^{k-1} \right) \right) \tag{25}$$

Iterative soft-thresholding algorithm (ISTA): special case with $h(x) = \|x\|_1$, i.e. Lasso problem.

7.4.1 Iterative soft-thresholding algorithm (ISTA)

$$
\begin{aligned}
\nabla \frac{1}{2}\|Ax-b\|^2 &= A^T(Ax-b) \\
x^{k+1} &= argmin_x \frac{1}{2}\|Ax-b\|^2 + (x-x^k)^T(A^TAx^k - A^Tb) + u\|x\|_1 + \frac{1}{2\lambda}\|x-x^*\|^2 \\
&= argmin_x \frac{1}{2\lambda}\left\|x - [x^k - \lambda(A^TAx^k - A^Tb)]\right\|^2 + \lambda\|\mu\|_1 \\
x_i^{k+1} &= sign[x^k - \lambda(A^TAx^k - A^Tb)]_i \left[\left|x^k - \lambda(A^TAx^k - A^Tb)\right|_i - \mu\right]_+
\end{aligned}
\tag{26}
$$

7.4.2 Fast iterative Shrinkage-Thresholding Algorithm

The Fast iterative shrinkage-thresholding algorithm's process is

$$
\begin{aligned}
x^k &= prox_{t_k, \nabla g(y)} \\
y &= x^k + \frac{t_k - 1}{t_{k+1}}(x^k - x^{k-1})
\end{aligned}
$$

Idea of the algorithm is calculate by GD, and then put the distance as penalty term into objective function.

7.4.3 Convergence properties

ISTA has a convergence rate of $O(\frac{1}{k})$

An accelerated version proposed in the same paper named FISTA

$$
\begin{aligned}
x^k &= \text{prox}_{\eta_k h}\left(y^{k-1} - \eta_k \nabla g\left(y^{k-1}\right)\right) \\
y^k &= x^k + \tfrac{t_k - 1}{t_{k+1}}\left(x^k - x^{k-1}\right)
\end{aligned}
\tag{27}
$$

Iteration complexity:

$$
f\left(x^k\right) - f\left(x^*\right) \leq \frac{f\left(x^1\right) - f\left(x^*\right) + \frac{L\left\|x^1 - x^*\right\|^2}{2}}{t_k^2} \leq \frac{4\left(f\left(x^1\right) - f\left(x^*\right)\right) + 2L\left\|x^1 - x^*\right\|^2}{(k+1)^2}
\tag{28}
$$

Example 7.4. Recall we want to solve

$$
\min F(X) := \mu\|X\|_* + \frac{1}{2}\|\mathcal{A}(X) - b\|^2
\tag{29}
$$

PGD gives that:

$$
\begin{aligned}
X^{k+1} &:= \arg\min_x \mu\|X\|_* + \left\langle g^k, X - X^k\right\rangle + \frac{1}{2\tau}\left\|X - X^k\right\|_F^2 \\
&= \arg\min_X \mu\|X\|_* + \frac{1}{2}\left\|X - \left(X^k - \tau g^k\right)\right\|_F^2
\end{aligned}
\tag{30}
$$

How to solve the sub-problem [2]

Given a matrix $Y \in \mathbb{R}^{m\times n}$, the sub-problem has the form:

$$
\min v\|X\|_* + \frac{1}{2}\|X - y\|^2
\tag{31}
$$

The optimal solution is

$$
X := S_\nu(Y) = U\,\text{Diag}\left(s_\nu(\sigma)\right)V^\top
\tag{32}
$$

$$\partial\|\boldsymbol{X}\|_* = \left\{\boldsymbol{U}\boldsymbol{V}^* + \boldsymbol{W} : \boldsymbol{W} \in \mathbb{R}^{n_1\times n_2}, \quad \boldsymbol{U}^*\boldsymbol{W} = 0, \quad \boldsymbol{W}\boldsymbol{V} = 0, \quad \|\boldsymbol{W}\|_2 \leq 1\right\} \tag{33}$$

$$\boldsymbol{Y} = \boldsymbol{U}_0\boldsymbol{\Sigma}_0\boldsymbol{V}_0^* + \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^* \tag{34}$$

Thresholding operator:

$$s_\nu(x) := \bar{x} \text{ with } \bar{x}_i = \begin{cases} x_i - \nu, & \text{if } x_i - \nu > 0 \\ 0, & \text{o.W.} \end{cases} \tag{35}$$

8 Stochastic optimization

$$\min_{x\in X} \left\{F(x) \equiv \mathbb{E}_\xi f(x,\xi)\right\} \tag{36}$$

Assume that

1. ξ is a random vector whose distribution P is supported on set $\epsilon \subset \mathcal{R}^d$
2. $f : X \times \epsilon \to \mathbb{R}$, and the exceptation
$$\mathbb{E}_\xi f(x,\xi) = \int_{\mathcal{E}} f(x,\xi)dP(\xi) \tag{37}$$
is well defined and has finite value for every $x \in X$
3. $f(\cdot,\xi)$ is convex for every $\xi \in E$

8.1 Sample average approximation

$$\min_{x\in X} \left\{\hat{F}_N(x) \equiv \frac{1}{N}\sum_{j=1}^{N} f\left(x,\xi_j\right)\right\} \tag{38}$$

8.2 Sample complexity:

suppose f has bounded variation, and let

$$V = \max\left\{f\left(x_1,\xi_1\right) - f\left(x_2,\xi_2\right) : x_1,x_2 \in X, \xi_1,\xi_2 \in E\right\} \tag{39}$$

then for any $\epsilon > 0$ and $\rho \in (0,1)$, sample size $N = [\frac{V^2}{2\epsilon^2}log(\frac{2}{\rho})]$ guarantees

$$\text{Prob}\left(\left|\hat{F}_N(x) - F(x)\right| \leq \epsilon\right) \geq 1-\rho, \forall x \in X \tag{40}$$

8.3 Stochastic gradient descent

$$x^{k+1} = x^k - t_k g\left(x^k, \xi_k\right) \tag{41}$$

Assumption: there exists a constant G such that

$$\mathbb{E}_\xi\left[\|g(x,\xi)\|_2^2\right] \le C^2, \forall x \in X \tag{42}$$

Fixed step size: $t = \frac{\theta D}{G\sqrt{N}}$ for some constant $\theta > 0$, then

$$\mathbb{E}\left[F\left(\tilde{x}^N\right) - F^*\right] \le \max\left\{\theta, \theta^{-1}\right\} \frac{DG}{\sqrt{N}} \tag{43}$$

Because of the existence of randomness, we need to put step size smaller to stabilize algorithm, so the convergences' speed is rational.

8.4 Convergence for strongly convex functions

Assume $F = \mathbb{E}_\xi f(x,\xi)$ is differentiable and strongly convex

$$F(y) \ge F(x) + \nabla F(x)^T (y - x) + \frac{\mu}{2}\|y - x\|_2^2 \forall x, y \in X \tag{44}$$

It holds that

$$\mathbb{E}\left[\left\|x^k - x^*\right\|_2^2\right] = r_k \le \frac{Q(\theta)}{k} \tag{45}$$

where$Q(\theta) = \max\left\{\theta^2 G^2 (2\mu\theta - 1)^{-1}, \left\|x^1 - x^*\right\|_2^2\right\}$,which is sub-linear.

If in addition ∇F is Lipschitz continuous with constant $L > 0$, then

$$\mathbb{E}\left[F\left(x^k\right) - F^*\right] \le \frac{L}{2}\mathbb{E}\left[\left\|x^k - x^*\right\|_2^2\right] \le \frac{LQ(\theta)}{2k} \tag{46}$$

Question: can we do better by exploiting finite average structure?

8.5 Stochastic variance reduced gradient(SVRG)

$$x^{k+1} = x^k - \eta\left(\nabla f_{i_k}\left(x^k\right) - \nabla f_{i_k}(\tilde{x}) + \nabla F(\tilde{x})\right) \tag{47}$$

The last term is all gradient but not mini-batch gradient, and update $\tilde{x}$ periodically (every few passes)

This is still a stochastic gradient method

$$\begin{aligned}
&\mathbb{E}\left[\nabla f_{i_k}\left(x^k\right) - \nabla f_{i_k}(\tilde{x}) + \nabla F(\tilde{x})\right] \\
=&\nabla F\left(x^k\right) - \nabla F(\tilde{x}) + \nabla F(\tilde{x}) \\
=&\nabla F\left(x^k\right)
\end{aligned} \tag{48}$$

The SVRG method convergence degree could be proven by setting $\tilde{x}^s = \frac{1}{m}\sum_{k=1}^m x^{k-1}$, but in practice using the last term to iterate can get faster speed.(Though No one can prove it temporarily.)

9 Convex Set Constrained Problem (FW method)

Let us consider a general form of a constrained problem:

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned} \tag{49}$$

where f is a linear and X is a ployhedron(linear programming).

or f is a convex function and X is a closed convex set (convex programming).

9.1 Conditional gradient method

$$\begin{aligned} \min \quad & f(x^k) + (\nabla f(x^k)^T (x - x^k)) \\ \text{s.t.} \quad & x \in X \end{aligned} \tag{50}$$

Use first-order approximation to replace objective function.

Then we have

1. Let the optimal solution be $\bar{x}^k$ and then let $d^k = \bar{x}^k - x^k$
2. if X is bounded, then d^k can always be found.
3. if ∇f is Lipschitz continuous, then any limit point of the iterates produced by Conditional Gradient Method is a stationary point.

9.2 Projection

Let $z \in X$. We call $\bar{z}$ the projection of z on X if $\bar{z}$ is optimal to the following problem

$$\begin{aligned} \min \quad & \tfrac{1}{2}\|x - z\|^2 \\ \text{s.t.} \quad & x \in X \end{aligned} \tag{51}$$

the projection is unique if X is convex.

Characterization of projection:

$$([x]_+ - x)^T \ (y - [x]_+) \geq 0 \tag{52}$$

A useful property of projection operator There is a useful property for projections:

$$\|[x]_+ - [y]_+\| \leq \|x - y\| \tag{53}$$

for all x and y.

Proof.

$$\begin{aligned} ([y]_+ - [x]_+)^\top (x - [x]_+) \leq 0 \\ ([x]_+ - [y]_+)^\top (y - [y]_+) \leq 0 \end{aligned} \tag{54}$$

From these relations we have

$$([y]_+ - [x]_+)^\top (x - [x]_+) + ([x]_+ - [y]_+)^\top (y - [y]_+) \leq 0 \tag{55}$$

then

$$\|[y]_+ - [x]_+\|^2 \leq ([y]_+ - [x]_+)^\top (y - x) \leq \|[y]_+ - [x]_+\| \cdot \|y - x\| \tag{56}$$

□

9.3 Gradient projection method

The basic idea is first to take a step along the gradient descent direction, and then project the point back into the feasible set:

$$\begin{aligned} \bar{x}^k &= [x^k - s_k \nabla f(x^k)]_+ \\ \bar{x}^{k+1} &= x^k + \alpha_k(\bar{x}^k - x^k) \end{aligned} \tag{57}$$

1. Iteration complexity(convex):$O(1/k)$
2. Linear convergence rate (strongly convex): $\left\|x^{k+1} - x^*\right\| \leq (1 - \frac{m}{M}) \left\|x^k - x^*\right\|$

Proof. Let us assume that $mI \preceq \nabla^2 f(x) \preceq MI$, then $I - s\nabla^2 f(x) \succeq 0$ for all x when $s \leq 1/M$. Futher let the step length $\alpha_k \equiv 1$.

Recall that we have $\|[y]_+ - [x]_+\| \leq \|y - x\|$

Then we have

$$\begin{aligned} &\left\|x^{k+1} - x^*\right\| \\ =&\left\|\left[x^k - s\nabla f\left(x^k\right)\right]_+ - [x^* - s\nabla f\left(x^*\right)]_+\right\| \\ \leq&\left\|x^k - x^* - s\left(\nabla f\left(x^k\right) - \nabla f\left(x^*\right)\right)\right\| \\ =&\left\| \int_0^1 \left(1 - s\nabla^2 f\left(x^k + t\left(x^* - x^k\right)\right) dt\right)\left(x^k - x^*\right)\right\| \\ \leq&\int_0^1 \left\|\left(I - s\nabla^2 f\left(x^k + t\left(x^* - x^k\right)\right)\right)\right\| dt \cdot \left\|x^k - x^*\right\| \end{aligned} \tag{58}$$

denote $\nabla f(x^k + t(x^* - x^k))$,then we have $\nabla^2 f(x^k + t(x^* - x^k))(x^* - x^k)$

Because

$$\begin{aligned} -mI \geq -\nabla^2 f(x^k + t(x^* - x^k)) \geq -MI \\ (1 - sM)I \leq I - s\nabla^2 f(x^k + t(x^* - x^k)) \leq (1 - sm)I \\ \left\|I - s\nabla^2 f(x^k + t(x^* - x^k))\right\| \leq (1 - \frac{m}{M})I \end{aligned} \tag{59}$$

The last term letting $s = 1/M$, and norm here is spectral norm. □

Example 9.1. Suppose $Q = Q^T$ and $Q \succ 0$. Quadratic optimization on the unit sphere

$$\begin{aligned} \min_x \quad & \tfrac{1}{2}x^\top Qx \\ \text{s.t.} \quad & \|x\| = 1 \end{aligned} \tag{60}$$

Then the projection is $[y]_+ = y/\|y\|$, objective gradient $\nabla f(x) = Qx$. Let step size in 57 $s_k = \alpha$ and $\alpha_k = 1$

Start form some $x^{(0)}$ satisfying $\left\|x^{(0)}\right\| = 1$

$$x^{(k+1)} = \frac{(1-\alpha Q)x^{(k)}}{\left\|(I-\alpha Q)x^{(k)}\right\|} \tag{61}$$

If $x^{(0)}$ is an eigenvector of Q, i.e. $Q_X^{(0)} = \lambda_0 x^{(0)}$,then

$$x^{(1)} = \frac{(I-\alpha Q)x^{(0)}}{\left\|(I-\alpha Q)x^{(0)}\right\|} = \frac{(I-\alpha\lambda_0)\,x^{(0)}}{\left\|(I-\alpha\lambda_0)\,x^{(0)}\right\|} = x^{(0)} \tag{62}$$

Unless $x^{(0)}$ corresponds to smallest eigenvalue, the algorithm is stuck at a non-optimal point.

Now, suppose $x^{(0)}$ is not an eigenvector of Q.

The eigenvectors of Q are $v_1, \ldots, v_n$, thus

$$(I-\alpha Q)v_i = (1-\alpha\lambda_i)v_i$$

Let $x^{(k)} = y_1^{(k)}v_1 + \cdots + y_n^{(k)}v_n$ and $\beta_k := \left\|(I-\alpha Q)x^{(k)}\right\|^{-1}$

Then

$$\begin{aligned} x^{(k+1)} &= \beta_k(I-\alpha Q)x^{(k)} \\ &= \beta_k(I-\alpha Q)\left(y_1^{(k)}v_1 + \cdots + y_n^{(k)}v_n\right) \\ &= \beta_k\left(y_1^{(k)}\,(1-\alpha\lambda_1)\,v_1 + \cdots + y_n^{(k)}\,(1-\alpha\lambda_n)\,v_n\right) \end{aligned}$$

Therefore,

$$y_i^{(k+1)} = \beta_k y_i^{(k)}\,(1-\alpha\lambda_i)$$

and thus

$$y_i^{(k)} = \beta_{k-1}y_i^{(k-1)}\,(1-\alpha\lambda_i) = \cdots = y_i^{(0)}\,(1-\alpha\lambda_i)^k\left(\prod_{j=0}^{k-1}\beta_j\right)$$

Rewrite for eliminating $\prod_{j=0}^{k-1}\beta_j$

$$x^{(k)} = y_1^{(k)}v_1 + \cdots + y_n^{(k)}v_n = y_1^{(k)}\left(v_1 + \sum_{i=2}^{n}\frac{y_i^{(k)}}{y_1^{(k)}}v_i\right)$$

Assume $y_1^{(0)} \neq 0$ we obtain

$$\frac{y_i^{(k)}}{y_1^{(k)}} = \frac{y_i^{(0)}\,(1-\alpha\lambda_i)^k\left(\prod_{j=0}^{k-1}\beta_j\right)}{y_1^{(0)}\,(1-\alpha\lambda_1)^k\left(\prod_{j=0}^{k-1}\beta_j\right)} = \frac{y_i^{(0)}}{y_1^{(0)}}\left(\frac{1-\alpha\lambda_i}{1-\alpha\lambda_1}\right)^k$$

if we assume $\lambda_1 < \lambda_i, i = 2, \ldots, n$

$$\frac{1-\alpha\lambda_i}{1-\alpha\lambda_1} < 1 \Rightarrow \frac{y_i^{(k)}}{y_1^{(k)}} \to 0 \tag{63}$$

which implies $x^{(k)} \to v_1$

Generally, we can start form a random $x^{(0)}$ so $y_1^{(0)} \neq 0$ with probability 1.

10 ADMM in linearly constrained problem

Consider the linearly constrained problem:

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

where $A \in \mathbb{R}^{p \times n}$

A natural thought is gradient projection method:

$$x^{k+1} := P_{\{x|Ax=b\}}\left(x^k - t_k \nabla f\left(x^k\right)\right) \tag{64}$$

But, the projection is not easy:

$$P_{\{x|Ax=b\}}(u) = \underset{\{x|Ax=b\}}{\arg\min} \|x-u\|_2^2 = u + A^\top \left(AA^\top\right)^{-1}(b - Au) \tag{65}$$

inexpensive only if $p \ll n$, or $AA^T = I$

10.1 Augmented Lagrangian method

For linearly constrained problem, augmented Lagrangian method (ALM) is usually used. Define augmented Lagrangian function:

$$\mathcal{L}_t(x;\lambda) := f(x) - \lambda^\top(Ax - b) + \frac{t}{2}\|Ax - b\|_2^2 \tag{66}$$

where λ is called the Lagrange multiplier or the dual variable.

The augmented Lagrangian method is (starting with $\lambda_0 = 0$)

$$\begin{cases} x^{k+1} := \arg\min\limits_x \mathcal{L}_t\left(x;\lambda^k\right) \\ \lambda^{k+1} := \lambda^k - t\left(Ax^{k+1} - b\right) \end{cases} \tag{67}$$

10.1.1 Convergence rate

Let $\bar{x}^k = \frac{1}{k}\sum_{j=1}^k x^j$. Then, for any fixed $\rho > 0$, we have

$$f\left(\bar{x}^k\right) - f\left(x^*\right) + \rho\left\|A\bar{x}^k - b\right\| \le \frac{\rho^2/t + \|\lambda^0\|/t}{k}$$

Example 10.1. Three variants of the l_1 minimization problem:

1. l_1-norm regularized problem (solved by)

$$\min \tau\|x\|_1 + \frac{1}{2}\|Ax - b\|_2^2 \tag{68}$$

   Use**proximal gradient method** to solve 1 (involves computing the prox operator of l1 norm)

2. l_1-constrained problem

$$\min \|Ax - b\|_2^2, \quad \text{s.t.} \quad \|x\|_1 \le \eta \tag{69}$$

   Use **gradient projection method** to solve 2 (involves computing the projection onto l1 ball)

3. Basis pursuit problem

$$\min \|x\|_1, \quad \text{s.t.} \quad Ax = b \tag{70}$$

Use **ALM** to solve 3 (involves inexactly minimizing the augmented Lagrangian function)

Advantage of Basis pursuit:

1. Basis pursuit problem requires the equality to hold (important for many applications)
2. ALM converges such that $Ax^k - b \to 0$ while the other models cannot guarantee this.

Usign ALM method to solve basis pursuit.

1. Specific iteration of Augmented Lagrangian function:

$$x^{k+1} = \arg\min_x \left(\|x\|_1 - \left\langle \lambda^k, Ax - b \right\rangle + \frac{t}{2} \|Ax - b\|_2^2 \right) \tag{71}$$

2. Update Lagrange multiplier(dual variable):

$$\lambda^{k+1} = \lambda^k - t \left(Ax^{k+1} - b \right) \tag{72}$$

Remark: check why here using proximal gradient method is irrational.

10.2 ADMM

$$\begin{aligned} \min \quad & f(x) + g(y) \\ \text{s.t.} \quad & Ax + By = b \end{aligned} \tag{73}$$

1. Both f and g closed and convex
2. Both f and g have special structures: easy proximal mappings
3. When f and g are non-smooth, the proximal gradient method can be used.
4. A special case is min f(x)+g(x), because it is equivalent to (by variable splitting)

$$\begin{aligned} \min \quad & f(x) + g(y) \\ \text{s.t.} \quad & x - y = 0 \end{aligned} \tag{74}$$

10.2.1 General form ofADMM

Augmented Lagrangian function

$$\mathcal{L}_t(x, y; \lambda) = f(x) + g(y) - \langle \lambda, Ax + By - b \rangle + \frac{t}{2} \|Ax + By - b\|_2^2 \tag{75}$$

ADMM:

$$\begin{cases} x^{k+1} = \arg\min_x \mathcal{L}_t \left(x, y^k; \lambda^k \right) \\ y^{k+1} = \arg\min_y \mathcal{L}_t \left(x^{k+1}, y; \frac{\lambda^k}{2} \right) \\ \lambda^{k+1} = \lambda^k - t \left(Ax^{k+1} + By^{k+1} - b \right) \end{cases} \tag{76}$$

Two subproblems:

$$
\begin{aligned}
x^{k+1} &= \arg\min_{x} f(x) + \frac{t}{2}\left\|Ax + By^k - b - \lambda^k/t\right\|_2^2 \\
y^{k+1} &= \arg\min_{y} g(y) + \frac{t}{2}\left\|Ax^{k+1} + By - b - \lambda^k/t\right\|_2^2
\end{aligned}
\tag{77}
$$

Solving Lasso by ADMM

$$
\begin{aligned}
&\min \tfrac{1}{2}\|Ax - b\|_2^2 + \lambda\|x\|_1 \\
&\min \tfrac{1}{2}\|Ax - b\|_2^2 + \lambda\|z\|_1 \\
&\text{s.t.} \quad x - z = 0 \\
&x^{k+1} := \left(A^\top A + tI\right)^{-1}\left(A^\top b + tz^k + y^k\right) \\
&z^{k+1} := S_{\lambda/t}\left(x^{k+1} - y^k/t\right) \\
&y^{k+1} := y^k - t\left(x^{k+1} - z^{k+1}\right)
\end{aligned}
\tag{78}
$$

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